

RISHI MULE

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EDUCATION

Northeastern University, Boston, MA **Dec 2025**
Master of Science in Computer Science, Khoury College of Computer Sciences GPA: 3.96/4.0
Coursework: Programming Design Paradigms, Database Systems, Algorithms, Cloud Computing, Machine Learning, MLOps
University of Mumbai, Mumbai, India **May 2023**
Bachelor of Engineering in Computer Engineering GPA: 3.88/4.0
Coursework: Data Structures, Object-Oriented Design, Analysis of Algorithms, Operating Systems, Big Data, NLP

EXPERIENCE

AI Engineer Intern | *CareEscapes AI, Cambridge, MA* **Feb 2026 – Present**

- Containerized 2 FastAPI inference services with Docker on Azure Container Apps and shipped React-side flows for instant quote display and clinic recommendation cards, cutting end-to-end deployment time from 40 to 12 minutes per release
- Assembled a cost-estimation module integrating procedure pricing, Amadeus flight APIs, and hotel APIs into a unified quote, holding predicted-versus-actual delta within plus-minus 12% across 35 sample patient itineraries
- Enforced HIPAA-aligned data handling on Azure with TLS 1.3, environment-based secrets management, and audit-logging hooks across 2 backend services, hardening the compliance posture for protected health information end-to-end

Software Developer Intern | *Thermobionics, Boston, MA* **Sep 2024 – Dec 2024**

- Built a full-stack React and Node.js analytics dashboard wired to a Python and Flask REST API on AWS to visualize live biometric data and reliably handle 10,000+ requests per minute under peak production load from connected hardware
- Implemented a Redis caching layer for high-frequency sensor readings, decreasing primary PostgreSQL write loads by 35% and dropping average query response time from 3.5 seconds to under 200 milliseconds in production
- Created automated CI/CD pipelines via GitHub Actions to run PyTest integration suites and deploy to AWS EC2, achieving 90% test coverage and cutting release cycles from weekly to daily across the engineering team

Software Engineer Intern | *InfoInnotronics Eng. & Tech, Pune, India* **Jun 2022 – Sep 2022**

- Architected a Spring Boot microservices backend coupled with an Apache Kafka event-streaming pipeline, ingesting real-time voltage and current metrics at 50,000+ events per second across distributed sensor nodes during peak loads
- Configured Jenkins CI/CD pipelines to automate regression testing for the energy monitoring software, catching 95% of integration bugs before the staging phase and saving 15+ engineering hours per week across the QA team
- Accelerated the core Python data ingestion script via multithreading and asyncio, increasing the maximum supported sensor node count from 50 to over 200 concurrent connections without additional infrastructure spend on the cluster

PROJECTS

Video Content Gap Analyzer (Hackathon Winner) | *FiftyOne, Twelve Labs, UMAP, Scikit-Learn* **Apr 2026**

- Engineered a FiftyOne plugin detecting video dataset gaps via Twelve Labs Marengo 512-d multimodal embeddings
- Tuned KMeans with silhouette scoring to flag sparse clusters and surface underrepresented categories at runtime
- Visualized cluster boundaries via UMAP 2D projection with grid-occupancy coverage scoring in the FiftyOne App

AI Code Review Agent | *LangGraph, Groq API, GitHub Actions, Pydantic, Langfuse, Python* **Feb 2026**

- Constructed a 5-node LangGraph pipeline parsing diffs and structuring findings via Pydantic schemas by severity
- Instrumented Langfuse observability tracking per-node execution time, token usage, and API cost across runs
- Benchmarked Llama 3.3 70B agent on synthetic PR diffs, hitting 68.2% precision and 50.0% recall on triage

ArXiv Intelligence MCP | *Model Context Protocol, ChromaDB, DocLing, RAG, ArXiv API* **Jan 2026**

- Designed a Model Context Protocol server querying ArXiv preprints via semantic RAG on ChromaDB vector embeddings
- Wired DocLing PDF ingestion to parse academic tables, formulas, and citations into chunked vector indexes

AI-GovBot (H.A.R.V.E.Y) | *LangChain, Kubernetes (GKE), Docker, Airflow, FAISS, MLflow, GCP* **Apr 2025**

- Deployed a RAG legal assistant via LangChain and FAISS with hybrid BM25 search, cutting hallucination rates by 20%
- Scaled containerized backend on Docker and Kubernetes (GKE), allocating peak-load resources with zero downtime
- Orchestrated Apache Airflow and TFDV pipelines validating legislative updates at a 99% data freshness SLA

OpenFold Mini-Implementation | *CUDA, NVIDIA Nsight Systems, PyTorch, C++, PyMOL* **Mar 2025**

- Parallelized attention matrix calculations via a custom CUDA kernel, accelerating inference 45% over CPU baseline
- Compressed the MSA database from 2 TB to under 100 GB via aggressive filtering, enabling consumer-GPU inference
- Profiled GPU memory bottlenecks with NVIDIA Nsight Systems, sustaining 80% tensor core utilization at peak load

Java Image Processing Application | *Swing, JUnit, MVC, Command & Strategy Patterns, Javadoc, Java* **May 2024**

- Applied Command and Strategy design patterns enabling 50-step undo/redo and pluggable filters across the engine
- Refactored convolution kernels with separable filters and multi-threading, dropping 1920x1080 runtime to 400 ms

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C/C++, SQL (PostgreSQL, MySQL), HTML/CSS, Bash, R, LaTeX
Frameworks: React, Node.js, Flask, Django, Spring Boot, FastAPI, LangChain, PyTorch, TensorFlow, Scikit-Learn
Cloud & Databases: AWS (S3, EC2, Lambda), GCP (BigQuery, GKE), Azure, PostgreSQL, MongoDB, Redis, FAISS
Developer Tools: Docker, Kubernetes, Git, GitHub Actions, Jenkins, Kafka, Airflow, Linux, MLflow, PyTest